Example:

$1000 instrument (principal protected note) consisting of

- 3-year zero-coupon bond with principal of $1000
- 3-year at-the-money call option on a stock portfolio currently worth $1000 (i.e. Strike price=$1,000)

The buyer of the note pays 1,000 today. At maturity (in 3 years), the buyer gets:

- $1,000 from the zero-coupon bond
- 0 from the call if the spot value of the portfolio is less than $1,000
- ST-1000 from the call if the spot value of the portfolio is more than $1,000

Example:

$1000 instrument (principal protected note) consisting of

- 3-year zero-coupon bond with principal of $1000
- 3-year at-the-money put option on a stock portfolio currently worth $1000 (i.e. Strike price=$1,000)

The buyer of the note pays 1,000 today. At maturity (in 3 years), the buyer gets:

- $1,000 from the zero-coupon bond
- 1000-ST from the put if the spot value of the portfolio is less than $1,000
- 0 from the put if the spot value of the portfolio is more than $1,000

PPN EXAMPLE

A bank decides to create a two-year principal-protected note on a non-dividend-paying stock by offering investors a zero-coupon bond with a face value of $100 plus a put option with K=$100. If the bank sets the cost of the note at $100 and interest rates are currently 8% per annum for all maturities, what is the maximum cost of the put option for the bank?

How will the answer change if interest rates were 4% instead?

$$Bond\ Price = \$100 * e^{-r*T} = \$100 * e^{-0.08*2} = \$85.2$$

$$p^{MAX} = \$100 - \$85.2 = \$14.8$$

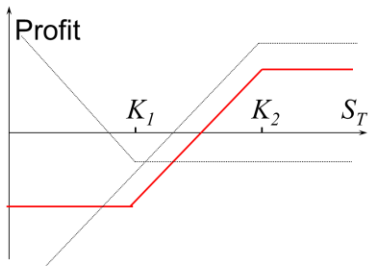
$$Bond\ Price = \$100 * e^{-r*T} = \$100 * e^{-0.04*2} = \$92.3$$

$$p^{MAX} = \$100 - \$92.3 = \$7.7$$

Problem 1.

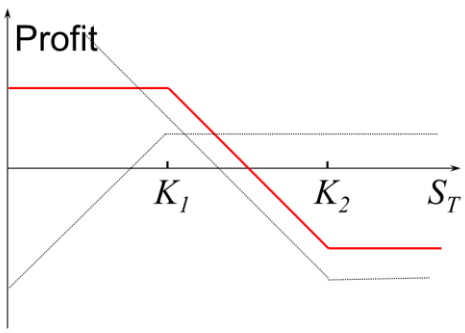
Suppose that put options on a stock with strike prices $30 and $35 cost $4 and $7, respectively. How can the options be used to create (a) a bull spread and (b) a bear spread? Construct a table that shows the profit and payoff for both spreads.

A bull spread is created by buying the $30 put and selling the $35 put. This strategy gives rise to an initial cash inflow of $3. The outcome is as follows:



Stock Price	Payoff	Profit	
$S_T \geq 35$	0	3	Nothing exercised
$30 \leq S_T < 35$	$S_T - 35$	$S_T - 32$	The sold put exercised
$S_T < 30$	St-35(short) +30-St(long)=-5	-5-(-3)=-2	Both exercised

A bear spread is created by selling the $30 put and buying the $35 put. This strategy costs $3 initially. The outcome is as follows

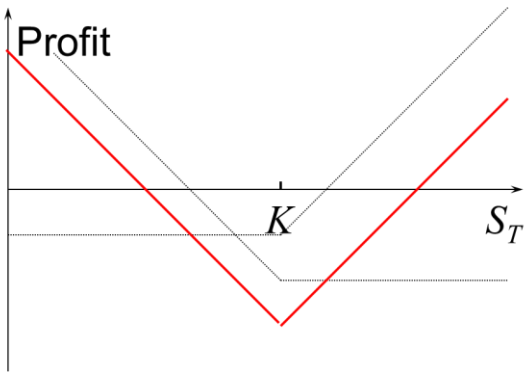


Stock Price	Payoff	Profit	
$S_T \geq 35$	0	-3	No exercise
$30 \leq S_T < 35$	$35 - S_T$	$32 - S_T$	The bought is exercised
$S_T < 30$	5 35-St+St-30=5	5-3=2	Both exercised

Problem 2.

A call with a strike price of $60 costs $6. A put with the same strike price and expiration date costs $4. Construct a table that shows the profit from a straddle. For what range of stock prices would the straddle lead to a loss?

A straddle is created by buying both the call and the put. This strategy costs $10. The profit/loss is shown in the following table:



Stock Price	Payoff	Profit	
$S_T > 60$	$S_T - 60$	$S_T - 70$	Exercise call
$S_T \leq 60$	$60 - S_T$	$50 - S_T$	Exercise put

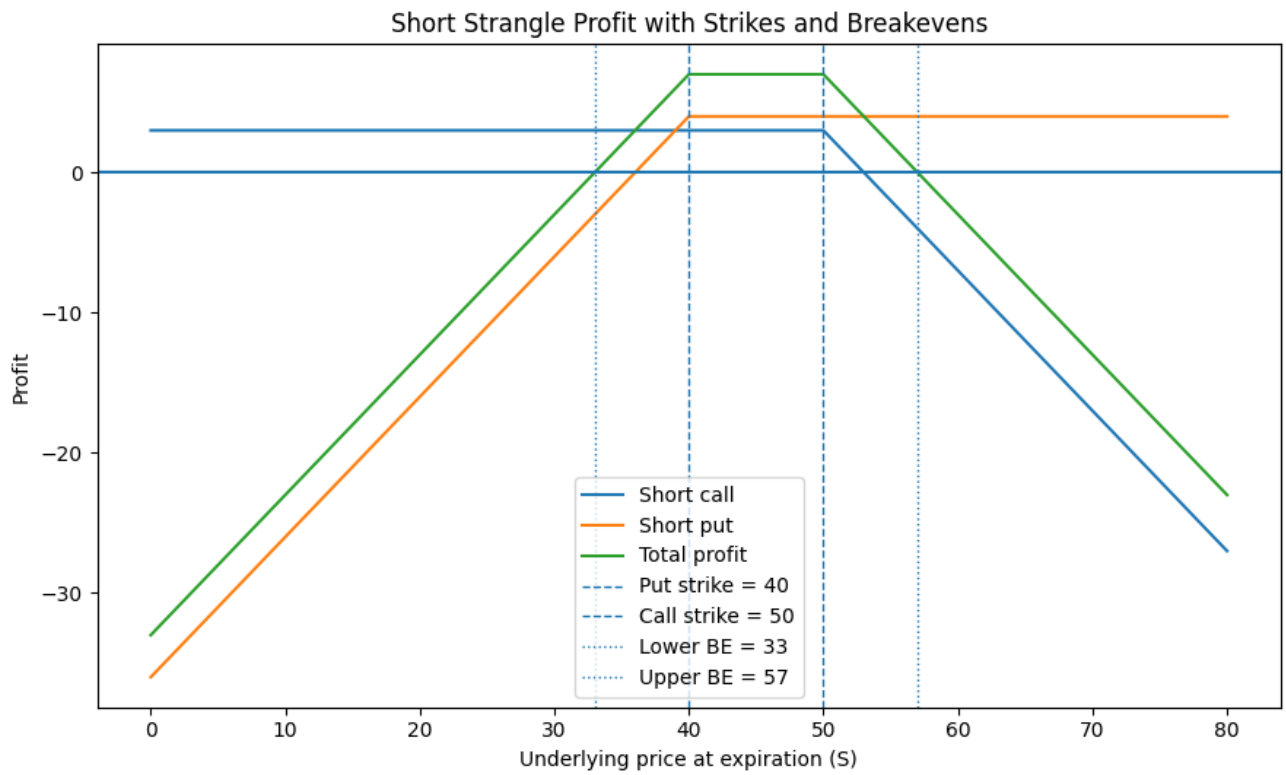
This shows that the straddle will lead to a loss if the final stock price is between $50 and $70.

Problem 3.

A trader sells a strangle by selling a call option with a strike price of $50 for $3 and selling a put option with a strike price of $40 for $4. For what range of prices of the underlying asset does the trader make a profit?

Seller gets $3+$4=$7

The trader makes a profit if the total payoff is less than $7. This happens when the price of the asset is between $33 and $57.



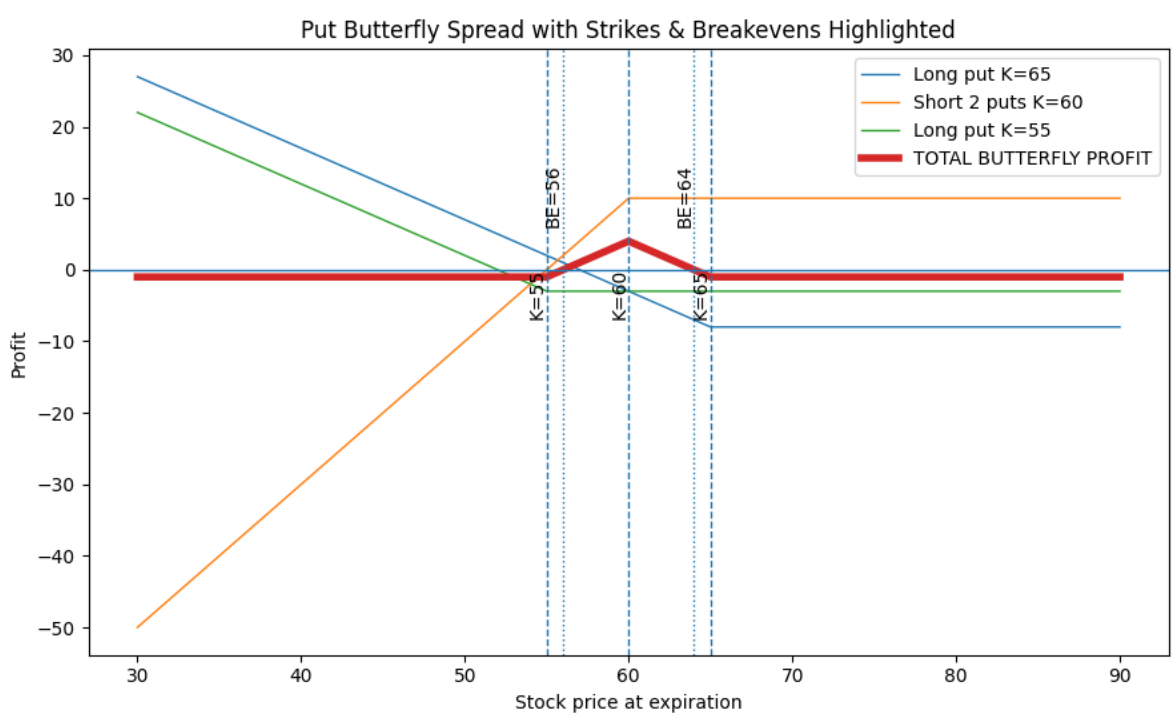
Problem 4.

Three put options on a stock have the same expiration date and strike prices of $55, $60, and $65. The market prices are $3, $5, and $8, respectively. Explain how a butterfly spread can be created. Construct a table showing the profit from the strategy. For what range of stock prices would the butterfly spread lead to a loss?

A butterfly spread is created by buying the $55 put, buying the $65 put and selling two of the $60 puts. This costs $3+8-2\times5=\$1$ initially. The following table shows the profit/loss from the strategy.

Stock Price	Payoff	Profit	What happens
$S_T \geq 65$	0	-1	Nothing exercised
$60 \leq S_T < 65$	$65 - S_T$	$64 - S_T$	K=65 exercised
$55 \leq S_T < 60$	65-S_T+2*(S_T-60) =S_T-55	S_T-55-1= S_T-56	K=65 and K=60 exercised
$S_T < 55$	65-S_T+2*(S_T-60)+55-S_T = 0	0-1=-1	All exercised

The butterfly spread leads to a loss when the final stock price is greater than $64 (65-1) or less than $56 (55+1).



Extra problem

An investor believes that there will be a big jump in a stock price, but is uncertain as to the direction. Identify six different strategies the investor can follow and explain the differences among them.

Possible strategies are:

Strangle

Straddle

Strip

Strap

Reverse calendar spread

Reverse butterfly spread

The strategies all provide positive profits when there are large stock price moves. A strangle is less expensive than a straddle, but requires a bigger move in the stock price in order to provide a positive profit. Strips and straps are more expensive than straddles but provide bigger profits in certain circumstances. A strip will provide a bigger profit when there is a large downward stock price move. A strap will provide a bigger profit when there is a large upward stock price move. In the case of strangles, straddles, strips and straps, the profit increases as the size of the stock price movement increases. By contrast in a reverse calendar spread and a reverse butterfly spread there is a maximum potential profit regardless of the size of the stock price movement.